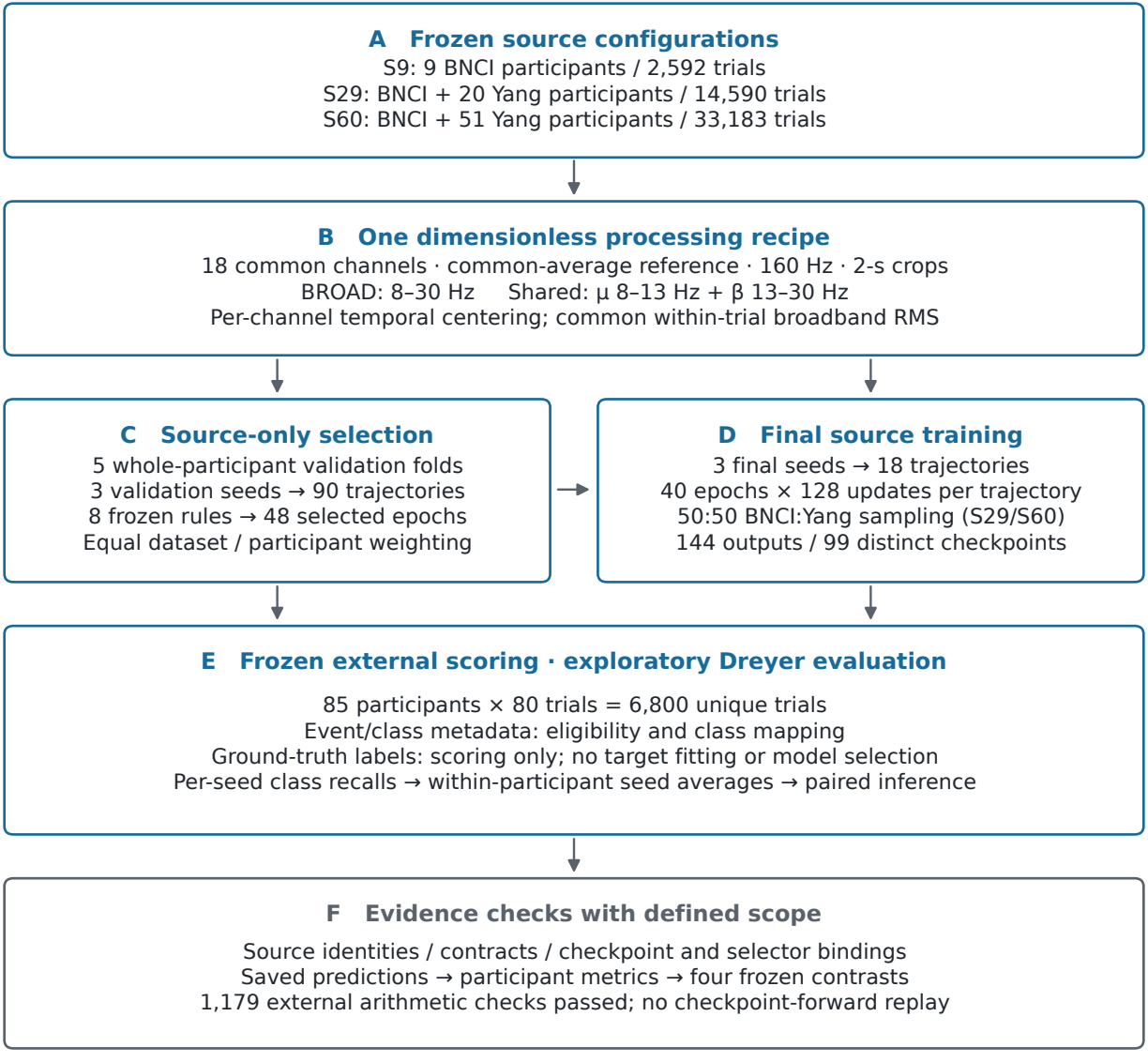
A Frozen source configurations
S9: 9 BNCI participants / 2,592 trials
S29: BNCI + 20 Yang participants / 14,590 trials
S60: BNCI + 51 Yang participants / 33,183 trials
B One dimensionless processing recipe
18 common channels · common-average reference · 160 Hz · 2-s crops
BROAD: 8–30 Hz Shared: μ 8–13 Hz + β 13–30 Hz
Per-channel temporal centering; common within-trial broadband RMS
C Source-only selection
5 whole-participant validation folds
3 validation seeds → 90 trajectories
8 frozen rules → 48 selected epochs
Equal dataset / participant weighting
D Final source training
3 final seeds → 18 trajectories
40 epochs × 128 updates per trajectory
50:50 BNCI:Yang sampling (S29/S60)
144 outputs / 99 distinct checkpoints
E Frozen external scoring · exploratory Dreyer evaluation
85 participants × 80 trials = 6,800 unique trials
Event/class metadata: eligibility and class mapping
Ground-truth labels: scoring only; no target fitting or model selection
Per-seed class recalls → within-participant seed averages → paired inference
F Evidence checks with defined scope
Source identities / contracts / checkpoint and selector bindings
Saved predictions → participant metrics → four frozen contrasts
1,179 external arithmetic checks passed; no checkpoint-forward replay